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AN SVD VIEW OF CONV-TASNET

Signal-processing structure, spectral dynamics, and low-rank compression

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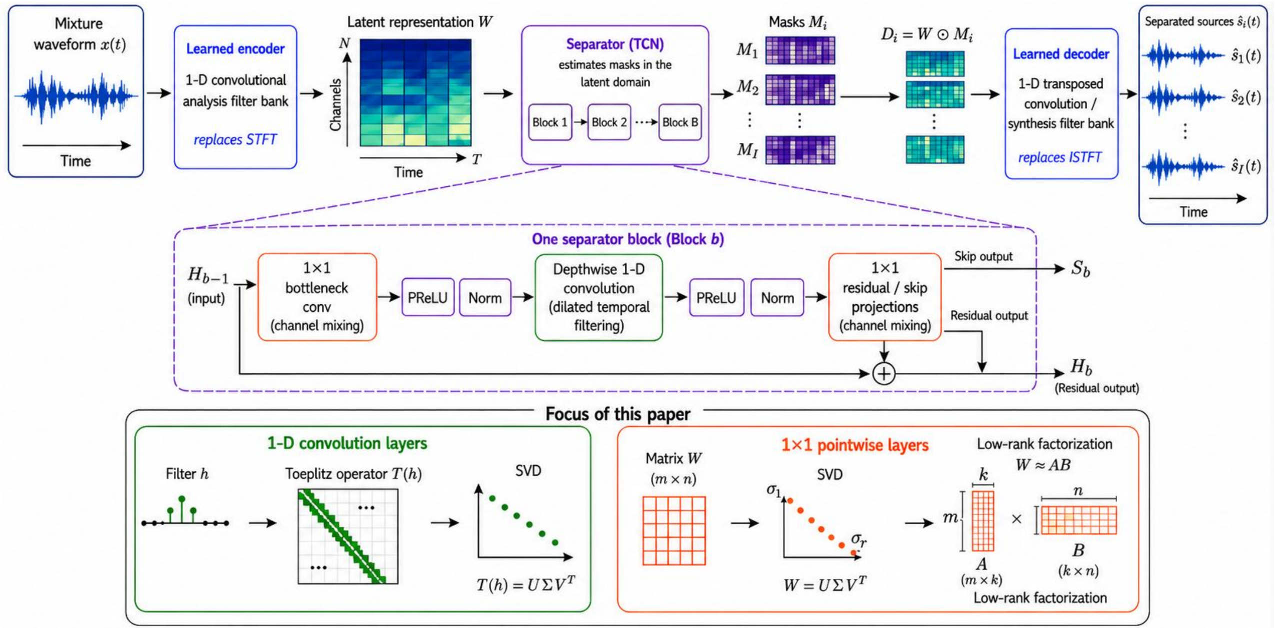
Conv-TasNet exposes two analyzable operator families



Study scope

Temporal operators

- Encoder/decoder filter bank
- Dilated depth-wise FIR filters
- Toeplitz interpretation: Szegő-type theory.

Pointwise operators

- Expand, residual, and skip maps in 24 TCN blocks (24 × 3 = 72)
- Bottleneck + I/O 72 + 2 = 74 matrices

Preserve temporal filtering; analyze and compress repeated channel mixing.

Two affine families require two SVD interpretations

Temporal convolution

A finite impulse-response filter becomes a rectangular Toeplitz operator. Its singular spectrum describes which temporal patterns are transmitted or suppressed.

$$y_n = \sum_{k=0}^{K-1} h_k\, x_{n+k}, \quad \mathbf{y} = T_N(h)\,\mathbf{x}.$$

Interpretation **Structured operator spectrum ↔ filter response**

Pointwise 1×1 convolution

At every time index, the layer applies the same explicit matrix to the channel vector. No tensor reshaping is needed for SVD.

$$\mathbf{y}_t = W\mathbf{z}_t, \qquad W \in \mathbb{R}^{m\times n}.$$

Interpretation **Direct singular directions ↔ channel-mixing modes**

Spectral concentration in a filter means selectivity; in a pointwise map it can also expose redundant channel directions.

Szegő-type theory: singular values to the filter response

$$M = N - K + 1, \qquad y = T_N(h)x, \qquad T_N(h) \in \mathbb{R}^{M \times N},$$

$$T_N(h) = \begin{bmatrix} h_0 & h_1 & \cdots & h_{K-1} & 0 & \cdots & 0 \\ 0 & h_0 & h_1 & \cdots & h_{K-1} & \ddots & \vdots \\ \vdots & \ddots & \ddots & \ddots & & \ddots & 0 \\ 0 & \cdots & 0 & h_0 & h_1 & \cdots & h_{K-1} \end{bmatrix}$$

$$G_N = T_N(h)^\mathsf{T} T_N(h), \quad \lambda_i(G_N) = \sigma_i(T_N(h))^2$$

$$\frac{1}{N}\sum_{i=1}^{N} g(\sigma_i^2) \longrightarrow \frac{1}{2\pi}\int_0^{2\pi} g\left(\left|H(e^{j\omega})\right|^2\right)\, d\omega.$$

$$\sigma_{\max}(T_N(h)) \longrightarrow \sup_{\omega} |H(e^{j\omega})|, \quad \sigma_{\min}(T_N(h)) \longrightarrow \inf_{\omega} |H(e^{j\omega})|$$

Distributional statement

As N grows, the empirical distribution of squared singular values $\sigma_i(T^N(h))^2$ approximates $|H(e^{j\omega})|^2$ asymptotically.

Extreme values

Large singular values correspond to temporal patterns that are strongly transmitted by the filter, while small singular values correspond to attenuated or near-null directions.

Scope boundary

The clean result assumes stride 1, no padding, and no dilation. Conv-TasNet’s dilation and encoder/decoder stride require sparse Toeplitz or polyphase extensions.

Two metrics: spectral concentration and subspace motion

Spectral concentration

Matrix dimension:

128×512 or 512×128

Entropy-based effective rank

$$p_i = \frac{\sigma_i}{\sum_j \sigma_j},$$

$$k_{\text{eff}}(W) = \exp\left(-\sum_i p_i \log p_i \right),$$

Subspace motion

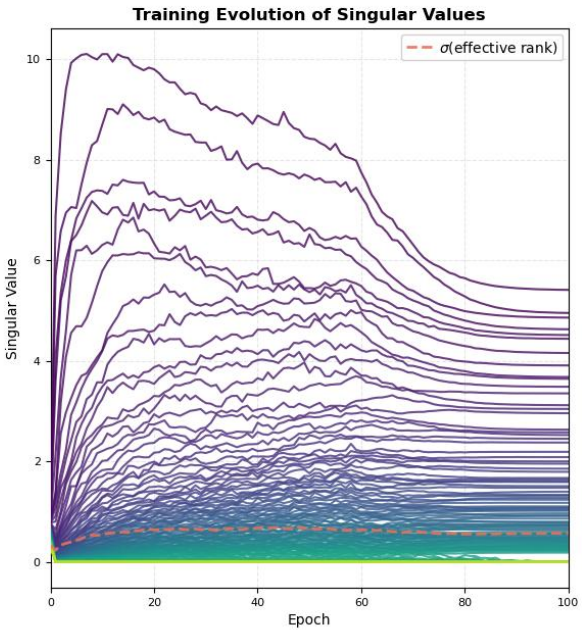
Principal angles compare the k-dimensional singular subspaces *Q* at consecutive epochs <*t-1*, *t*>. A small Grassmann geodesic distance d_G means the learned orientation has stabilized.

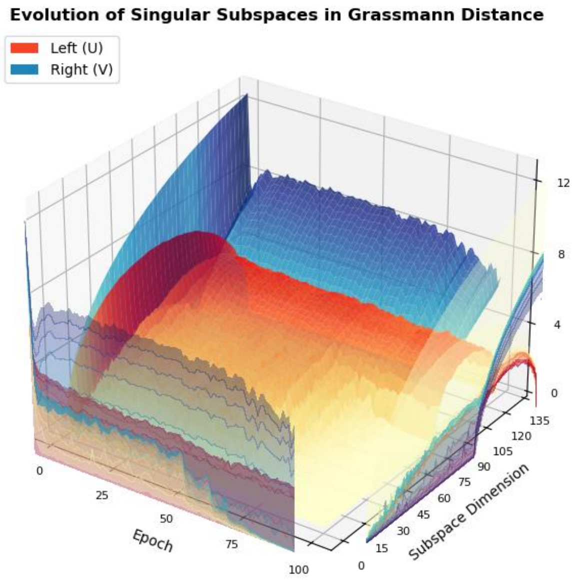
$$\cos\theta_i = \sigma_i(Q_t^{\mathsf{T}} Q_{t-1})$$

$$d_{\mathrm{G}}(Q_t, Q_{t-1}) = \left(\sum_i \theta_i^2 \right)^{1/2}.$$

Singular values measure gain; principal angles measure rotation; effective rank summarizes concentration.

Dominant singular directions form early, then stabilize





Three observations

1 **Early stratification**: Leading values separate within a few epochs; most of the tail remains weak.

2 **Subspace stabilization**: Dominant singular subspaces move strongly at the start, then change slowly.

3 **Compression chance**: The tail is not zero, but much of the useful map is carried by fewer directions.

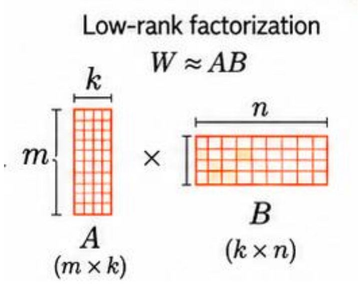
Effective rank becomes an operational boundary between dominant directions and weaker tail directions.

Truncated SVD creates a shape-dependent bottleneck

$$W_k = U_k \Sigma_k V_k^{\mathsf{T}} = AB,$$

$$A = U_k \Sigma_k^{1/2}, \quad B = \Sigma_k^{1/2} V_k^{\mathsf{T}},$$

$$k(m+n) < mn \iff k < \frac{mn}{m+n}.$$



Parameter economics

Original matrix:
***mn* parameters**

Two factors
***k(m+n)* parameters**

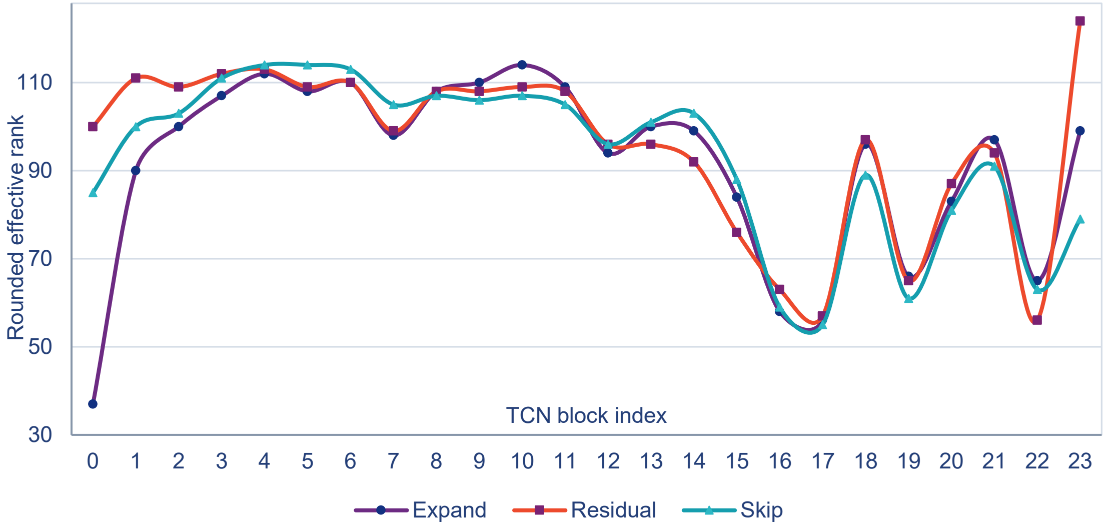
128×512 map
break-even: *k* < 102.4

What changes All 74 pointwise maps become two factors with an internal rank-*k* bottleneck.

What remains Temporal filters, encoder, decoder, masking, and the training recipe.

A rank can improve approximation yet expand a layer if it exceeds the shape-dependent break-even threshold.

Truncation at Effective Ranks



Parameter accounting

Component	Parameters
Pointwise before	4,915,200
Pointwise after	4,441,600
Removed	**473,600**
Whole model	5,524,145 → 5,050,545
Reduction	8.57% · 1.094× ≈ 1.10×

The rank profile is non-monotonic across depth; one global bottleneck would miss this structure.

EXPERIMENTS

One backbone, three regimes, one metric

Backbone Non-causal Conv-TasNet · N=512 · L=16 · B=128 · H=512 · P=3 · X=8 · R=3

Data Libri2Mix · 8 kHz · min mode · train (33.6k, 4-s segments), dev (3.8k), and test (4k) splits

VALIDATION OBJECTIVE

$$\mathcal{L}_{\mathrm{val}} = -\,\mathrm{SI{-}SNR} \qquad \text{(lower is better)}.$$

POST HOC FINE-TUNING

Best full-model checkpoint
→ SVD truncation
→ fine-tune

EARLY CHECKPOINT

Factorize at epoch 1, 5, or 10
→ continue training

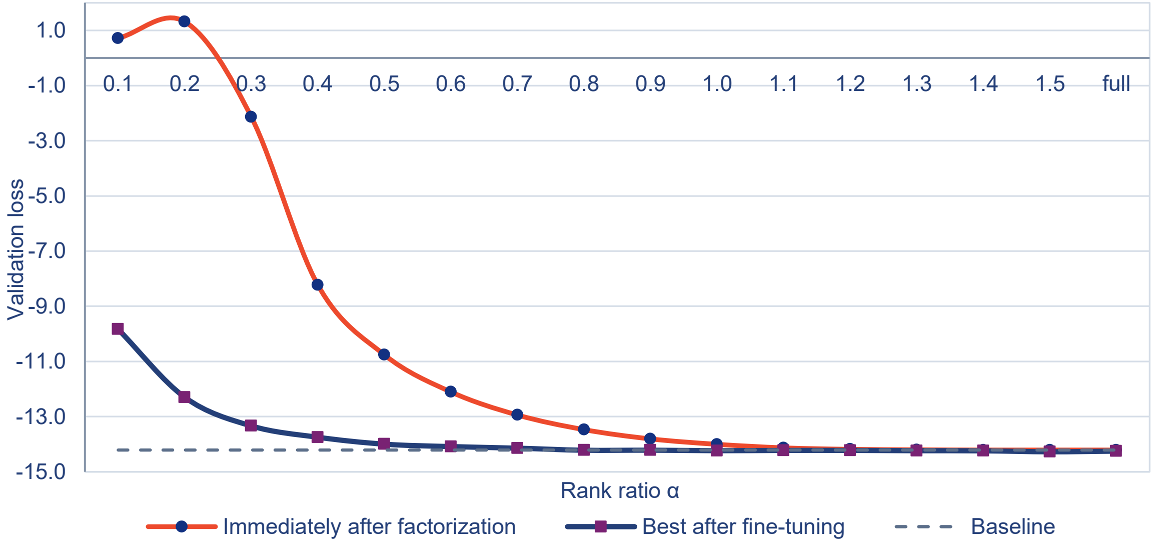
RANDOM INITIALIZATION

Choose a rank schedule
→ initialize A and B randomly
→ train from scratch

All regimes: Adam · 100 epochs · 4-s segments · LR halved on validation plateau · gradient clipping

EXPERIMENT 1

Post-hoc recovery becomes reliable near effective rank



Transition region

α = 0.8-1.0
1.37×–1.10× compression
Best loss reaches
−14.2181 to −14.2339
Baseline loss: -14.2096

Aggressive truncation

Aggressive truncation hurts immediately, but fine-tuning recovers much of the loss.

α > 1 Loss gains are tiny while compression disappears.

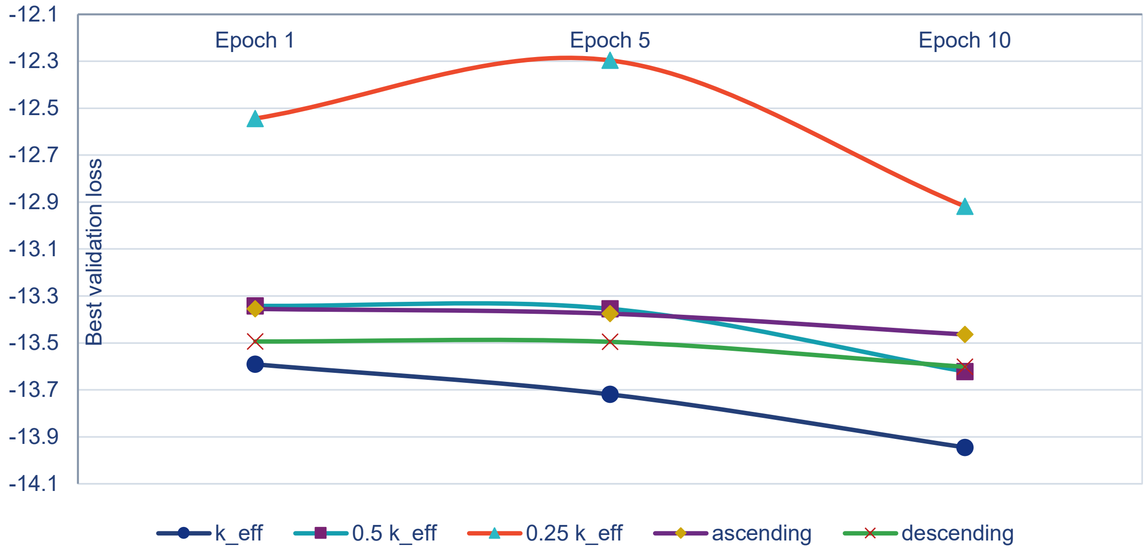
Each layer receives its own rank; α controls truncation around that layer's effective scale: $k_\ell \propto \alpha \cdot k\_eff(W_\ell)$

Complete post-hoc results: all 16 rank ratios

α	Compression	Initial loss	Best after FT	Δ from baseline	Best epoch
Baseline	1.00×	—	−14.2096	0.0000	78
0.1	8.71×	0.7177	−9.8351	4.3745	+56
0.2	4.93×	1.3163	−12.2977	1.9119	+51
0.3	3.44×	−2.1353	−13.3362	0.8734	+52
0.4	2.64×	−8.2283	−13.7470	0.4626	+38
0.5	2.14×	−10.7592	−13.9976	0.2120	+38
0.6	1.80×	−12.1045	−14.0856	0.1240	+15
0.7	1.56×	−12.9411	−14.1430	0.0666	+25
0.8	1.37×	−13.4710	−14.2181	−0.0085	+19
0.9	1.22×	−13.8072	−14.2162	−0.0066	+11
1.0	**1.10×**	**−14.0033**	**−14.2339**	**−0.0243**	**+7**
1.1	1.01×	−14.1314	−14.2247	−0.0151	+13
1.2	0.94×	−14.1845	−14.2241	−0.0145	+9
1.3	0.90×	−14.2017	−14.2350	−0.0254	+19
1.4	0.88×	−14.2096	−14.2390	−0.0294	+15
1.5	**0.86×**	**−14.2101**	**−14.2742**	**−0.0646**	**+13**
full	0.81×	−14.2097	−14.2507	−0.0411	+16

Highlighted: α=1.0 is the practical compression point; α=1.5 is the best-loss but over-parameterized point.

Later factorization preserves stronger learned subspaces



Consistent ordering

- Effective rank outperforms smaller rank rules at epochs 1, 5, and 10.
- Every schedule improves when factorization is delayed.
- Descending beats ascending and converges earlier.

Interpretation

Later checkpoints contain more mature dominant subspaces.

Factorization can be introduced early, but the model pays a quality cost when the useful subspaces are still forming.

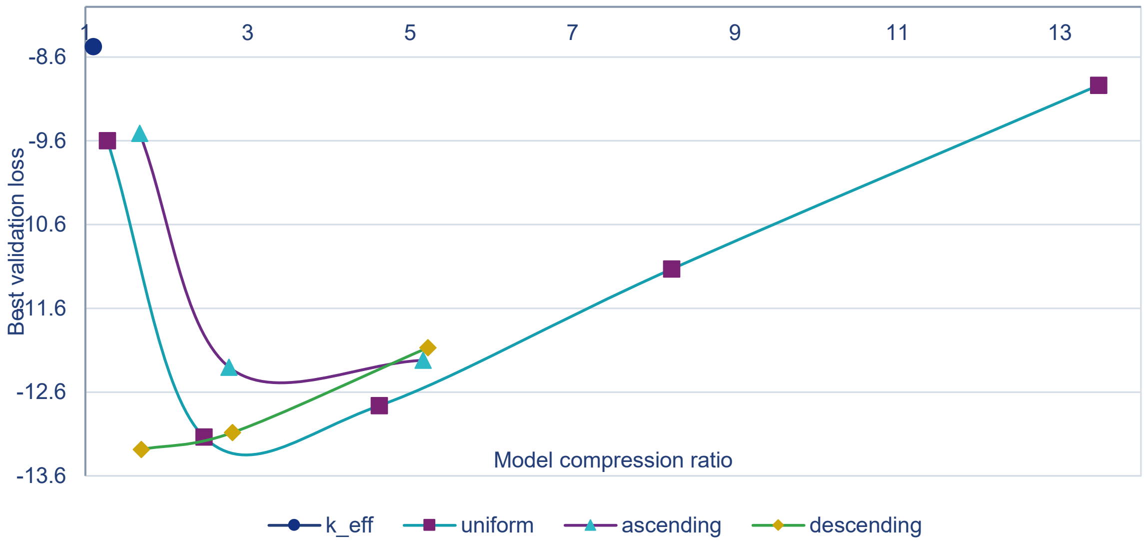
All 15 checkpoint-factorization runs

Factorize	Rank rule	Compression	Checkpoint	Initial	Best loss	Best epoch
Epoch 10	**k_eff**	**1.24×**	**−10.8078**	**−10.7734**	**−13.9449**	**+81**
Epoch 10	0.5 k_eff	2.39×	−10.8078	−10.1671	−13.6216	+83
Epoch 10	0.25 k_eff	4.49×	−10.8078	−7.7510	−12.9179	+69
Epoch 10	ascending	2.06×	−10.8078	−9.2964	−13.4640	+93
Epoch 10	descending	1.98×	−10.8078	−9.7609	−13.6017	+73
Epoch 5	**k_eff**	**1.34×**	**−9.6479**	**−9.6290**	**−13.7195**	**+88**
Epoch 5	0.5 k_eff	2.59×	−9.6479	−9.3568	−13.3547	+81
Epoch 5	0.25 k_eff	4.84×	−9.6479	−7.8804	−12.2960	+85
Epoch 5	ascending	2.24×	−9.6479	−9.2429	−13.3755	+95
Epoch 5	descending	2.12×	−9.6479	−9.0872	−13.4953	+76
Epoch 1	**k_eff**	**1.39×**	**−5.8962**	**−5.8926**	**−13.5912**	**+95**
Epoch 1	0.5 k_eff	2.68×	−5.8962	−5.8578	−13.3419	+89
Epoch 1	0.25 k_eff	4.99×	−5.8962	−5.7438	−12.5440	+85
Epoch 1	ascending	2.37×	−5.8962	−5.8610	−13.3551	+96
Epoch 1	descending	2.17×	−5.8962	−5.8267	−13.4938	+76

Highlighted rows are the k_eff rule at each checkpoint; the epoch-10 run is the overall best early-factorization model.

EXPERIMENT 3

From scratch, simple rank schedules work better



Best from scratch

Linear 100→20
−13.2841 at 1.69×

Uniform sweet spot

Uniform 40
−13.1355 at 2.46×

Observations

Higher-rank rules don't give better performance – contradictory to the trend in pre-trained cases!

Moderate rank constraints perform best from random initialization

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All 12 random-initialization rank schedules

Rank schedule	Family	Compression	Best loss	Best epoch
Trained k_eff profile	k_eff	1.10×	−8.4764	100
Uniform 80	uniform	1.27×	−9.5967	95
Uniform 40	**uniform**	**2.46×**	**−13.1355**	**97**
Uniform 20	uniform	4.62×	−12.7622	96
Uniform 10	uniform	8.22×	−11.1319	81
Uniform 5	uniform	13.48×	−8.9363	94
Linear 20→100	ascending	1.67×	−9.5110	100
Linear 100→20	**descending**	**1.69×**	**−13.2841**	**97**
Linear 10→60	ascending	2.77×	−12.3025	95
Linear 60→10	descending	2.81×	−13.0842	85
Linear 5→30	ascending	5.16×	−12.2210	99
Linear 30→5	descending	5.22×	−12.0690	99

Highlighted: Uniform 40 offers the strongest compression/quality balance; Linear 100→20 gives the best loss.

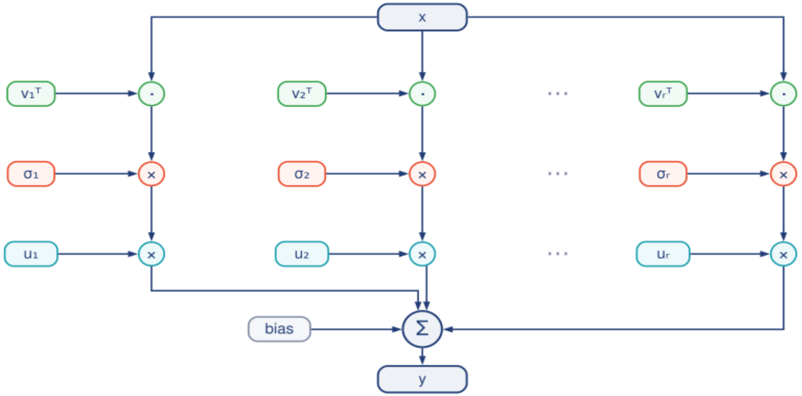
The evidence supports compression, not yet acceleration

ESTABLISHED

- **SVD gives two complementary readings:** Temporal filters can be viewed through Toeplitz/Szegő theory; pointwise maps reveal subspace dynamics.
- **Low-rank approximation is broadly applicable**: Convergence is shown in factored post-hoc fine-tuning, few-epoch pretraining, and random init.

NOT YET ESTABLISHED

- Hardware latency, throughput, memory traffic, or energy savings.
- Full Toeplitz theory for dilation, stride, padding, overlap-add, and polyphase structure.



SORO (Sum of Rank One) Layer

Next experiments

1 **Hardware: latency, throughput, and energy**
2 **Training: adaptive rank selection**
3 **Theory: dilation, stride, and polyphase structure**

Parameter compression is necessary evidence for efficiency—but not sufficient evidence for acceleration.

Three design rules for low-rank Conv-TasNet

1 **Post-hoc Fine-tuning**

Aggressive truncation hurts immediately, but fine-tuning recovers much of the loss; fine-tuning can slightly outperform the best model baseline.

2 **Pre-conditioned Training**

A slight whole-model warm-up plus low-rank training can recover much of the performance.

3 **Training from Scratch**

Moderate rank constraints perform best for factored, random-initialized Conv-TasNet.

Numerical anchors

−14.2096
full-model baseline

−13.2841
best from scratch · 1.69×

−14.2339
post hoc at effective rank · 1.10×

Code + results

github.com/Jovie-Liu/Low-Rank-Conv-TasNet

Questions